serves as a lipophilic group to move the compound through the cell's semi-permeable membrane and block metal-binding sites within microorganism enzymes. The latter was attributed to the fact that complexes may deactivate different cellular enzymes, which play a vital role in the microorganism metabolic pathways. It was also suggested that the toxicant's ultimate action was the denaturation of one or more cell proteins, impairing normal cellular processes as a consequence. Other factors that also increased the antimicrobial activity were the solubility, conductivity and length of bond between metal and ligand.

A comparative study of Schiff base and its complex indicate that the metal complexes exhibit higher antibacterial activity than the free ligands.[4,8] This is probably due to the greater lipophilic nature of the complex. Such increased activity of the metal chelets can be explained on the basis of overtones concept and chelation theory. According to overtones concept of cell permeability the lipid membrane that surrounds the cell favours the passage of only lipid soluble materials due to which liposolubility is an important factor which controls the antimicrobial activity. On chelation the polarity of the metal ion will be reduced to a greater extent due to the overlap of the ligand orbital and partial sharing of positive charge of metal ion with donor groups.

RESULT & DISCUSSION

The expected broad peak for O-H stretching (H-bonded) at 3294.94 cm^{-1}, bidendate carboxylic acid functioning as ligand at 1489.05.cm, C-O stretching at 1230.58 cm^{-1}, O-C stretching at 1369.46 cm^{-1} and M-O stretching at 680.66 cm^{-1} are present in the curve. Similarly, the loss pattern is just what we expect for the formulation as shown. We observe that the FTIR curves (table -3) of sample R3 contain almost all the peaks which are expected for its formulation. We observe that the FTIR curves contain almost all the peaks which are expected for its Formulation $Cr_2O.CH_3COOH.CH_3COCOOH.2H_2O$. It is supported by the peaks at 3298.28cm^{-1} (O-H stretching), 964.41cm (C-O-H stretching) 1489.05 cm^{-1} (bidentate carbxylate ligand), 844.82 cm^{-1} (-COOH co–ordinated), 964.41.cm-1 (C-O stretching) etc. Also the loss pattern in TGA-DTA curve supports the proposed formulation. The first stage loss between the temperature ranges 31.98-140^oC corresponds to the fragmentation of CH_3COOH. The second stage loss of lattice water. The substrate might have been co- ordinated through two COOH groups, as the loss is taking at higher temperature.

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Effects of global warming & climate change- Opinions, thoughts and remedial strategies

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Abstract- Squandering of precious natural resources, burning of fossil fuels & release of CO_2 in the atmosphere acting as glass panel not allowing the heat reradiation in the outer space, suicidal pollution of rivers, lakes & oceans, unplanned construction activities, and Dam construction have caused immense destruction to biosphere resulting into global warming & climate change. The impact of climate change on the loss of biodiversity (the latter is critical in bringing resilience impacting fertility) is so great that time is not far when world would face an ecological collapse & if the worst came to the worst, the extinction of life. It is true that fossil fuel (coal), power the world economy yet if the consumption of coal, oil, gas continues unabated the environmental problem won't stop. The increasing population needs, forced humans to indulge in deforestation, killing wild animals, exploit resources, growth of industries, excessive use of pesticides, insecticide, fertilizes, use of coal for energy all have destroyed environment immensely. That an increase of temperature of 1.5 Celsius over pre - industrial level would substantially threaten life as the world would see heat wave, heavy rainfall, extreme weather conditions, shortage of water, reduced farm output, bleaching of corals, rise of sea level & many more events. The author have incorporated thoughts and opinions of different authors at one platform, discussed the consequences and have suggested certain remedial measures.

Key words: Global warming & climate change, consequences, thoughts & opinions remedial measures.

INTRODUCTION

The earth's biosphere is a complex system in which events and crisis occur which is indicated by non-linear unpredictable changes. The increasing CO_2 levels leads to melting of ice caps and increase in sea levels. This in turn would disrupt life of people from Islands of Maldives, Marshall Islands, Nauru and Tuvalu and other coastal low lying areas, their histories, traditions and cultures will be lost forever.

Climate change impacts insects and pests. The decline of a parasitoid, wasps and flies playing an important role in biological pest control has been observed at various places like Nepal (decline of natural enemies of pests) United States (decline of nearly 40% grassland bird index during 1968 and 2014). In addition to these extreme weather events cause disruption to species distribution and yields. Cold and windy days in spring disrupts pollination, rising temperature can shift the range of fishes impacting fisheries. Briefly understanding all these changes, the present authors have concentrated on what are the present scenario of global warming, its consequences and possible remedial measures. The present article includes thoughts and opinions of different authors who wrote on global warming.

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MATERIALS & METHODS

Different research papers and informations have been consulted and incorporated.

Present scenario

Pollution at glance

Oxygen in the atmosphere is an absolute necessity to live comfortably but nature has been exploited for acquiring comforts thus air has become polluted. In Delhi itself "the number of four wheelers have exceeded one crore. These vehicles emit CO_2, CO, SO_2, nitrogen hydrocarbons, etc in the air of urban areas, is now polluted.[1]

Industrialization has played a great role in polluting the air, smoke coming out of chimneys contain lead, mercury, copper, cadmium, arsenic & asbestos etc. Beside these, it also emits CO_2, CO, SO_2, H_2S hydrogen fluoride. Petroleum refinery too releases SO_2 & nitrogen oxide.

In many places, burning of solid biomass in poorly ventilated spaces add to environmental pollution. Nearly 1.3 million premature deaths in India have been reported by WHO – 2018. It also leads to approximately 30% of ambient air pollution in India. "One of the major constituents of house hold air pollution is PM 2.5 which affects local & regional climate change".[2]

The present authors opine that in order to reduce such pollution it would be worth to provide liquefied petroleum gas cylinders to each & every house hold. Ujjwala Yojana of the Govt. of India is an effort towards this venture.

Climate change resulting from growing green house gas (GHG) emissions from human (anthropogenic) activities is the prime cause of global warming and if the CHG emission is not controlled, the temperature of the earth is bound to increase. The affect of temperature increase can have potentially harmful effect on our ecosystems, biodiversity health, behaviour & livelihood, also many other events can happen.

Green house gas CO_2 is released in the atmosphere due to burning of fossil fuels (coal, oil, natural gas) etc or via domestic cooking. Apart from emissions from transport, industrial activity, changes in land use, agriculture (including live stock) & waste management are also responsible. Other short lived climate pollutants include black carbon, methane and ground level ozone, which along with other pollutants, particulate matters from the same sectors combine to aggravate air quality & cause climate change.[3]

Rajendran (2019)[4] observed that India's emission of CO_2 are low at 1.8 tonnes per capita, compared to the world average of 4.2 tonnes possibly due to reduction in the consumption of petroleum products and the decline in Cement production.

CONSEQUENCES

Health Hazards:-

In areas covered by snow such as Arctic & Himalayan region, the deposition of Black carbon onto ice & snow darkens the surface, increasing the absorption of solar radiation.[5] Higher temperature raises the level of ozone and other pollutants in the air which causes cardiovascular diseases.

In all probability it raises the level of nitrogen oxide (NOX), ozone (O_3), black carbon (BC), particulate matters, dust & floating pollen. These are pollutants to which people are exposed.

Dr. Nayar, HOD Respiratory Medicare Allergy and Sleep disorders at BLK Super - specialty Hospital opines that particles less than 2.5 can directly enter our respiratory system and can lead to breathlessness, cough, fever and even choking. Our nervous system gets affected & can experience headache, nausea or vomiting. It also affects heart.

Ozone

Ozone is a ubiquitous urban pollutant (Dobson unit is used for measuring the density of ozone; one Dobson unit = a layer 0.01 mm thin layer at 0-1 ATM (Atmospheric Pressure). It has been found that CO_2 is the main culprit of thinning of ozone layer. In human, inhalation of ozone in amounts low as 0.1 to 0.2 ppm is sufficient to cause toxicity characterized by edema, airways hyper – responsiveness, alveolar epithelium damage & impaired host defense.[6]

Acute inhalation of ozone is associated with an inflammatory response characterized by the accumulation of macrophages at the site of injury. Thus cells along with resident alveolar epithelial cells get activated & release cytotoxic & pro – inflammatory mediators.

Ozone & Agriculture

Surface ozone is generated by chemical reactions between primary pollutants such as oxides of nitrogen and volatile organic compounds in the presence of sunlight.